

Autonomous Articulated Vehicle Control via Deep Reinforcement Learning and High-Dimensional Perception

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Chapter 1

Introduction

1.1 Motivation

The increasing demand for autonomous logistics and the growth of large-scale goods transportation have positioned articulated tractor-trailer vehicles as a critical target for autonomous control research. The transition from rigid-body autonomous navigation to the control of articulated tractor-trailer systems represents a significant escalation in mechanical complexity, perception requirements, and control instability, particularly during reverse maneuvering.

Unlike single-body vehicles, tractor-trailer systems introduce non-holonomic constraints at the hitch point that fundamentally alter the stability characteristics of the vehicle. In reverse motion, the articulation angle γ between the tractor and trailer exhibits open-loop instability: absent precise closed-loop control, any perturbation grows unbounded, culminating in the well-known jackknife condition from which no steering correction can recover the trailer’s alignment. Classical approaches such as Nonlinear Model Predictive Control (NMPC) can manage this instability by optimising over a prediction horizon, but their computational demands and reliance on accurate system models limit their applicability in unstructured or sensor-rich environments.

1.2 Problem Definition

The core challenge addressed in this thesis is twofold. First, the kinematic and dynamic complexity of tractor-trailer systems makes it difficult to design reward functions and control policies that simultaneously optimise path tracking performance and hitch stability. Second, real-world autonomous deployment requires perception of lane boundaries and obstacles directly from high-dimensional sensor data, rather than from pre-computed symbolic state representations.

Prior work in the ME 780 project evaluated Deep Deterministic Policy Gradient

(DDPG) and Decision Transformer (DT) models for path following in single-vehicle systems. While DDPG achieved cross-track errors competitive with PID control, both learning-based methods showed significant degradation under novel conditions, and neither was tested on articulated dynamics or reverse motion.

1.3 Thesis Contributions

This thesis extends the ME 780 foundational work by:

- developing a full kinematic and dynamic model of a tractor-trailer system including hitch instability analysis;
- designing a vision-based RL observation space using CNN encoders and autoencoders operating on bird’s-eye-view (BEV) occupancy grids;
- engineering a reward function that balances cross-track error, heading error, hitch articulation stability, and collision avoidance;
- implementing and benchmarking DDPG and Decision Transformer agents against PID and NMPC baselines across forward, reverse, and obstacle-avoidance tasks;
- validating the trained policies in the Gazebo simulation environment and transferring to the Agilex hardware platform via ROS2 and Autoware.

The research aligns with the objectives of the Mechatronics Vehicle Systems (MVS) Lab and the WATonoBus project at the University of Waterloo.

1.4 Institutional Context

This work is submitted in partial fulfilment of the requirements for a Master of Applied Science (MASc) degree in the Department of Mechanical and Mechatronics Engineering at the University of Waterloo. The degree requires six terms of full-time registration, completion of 3–5 graduate-level courses, appointment of two additional faculty readers, and a mandatory 15-day public display period. The target completion date is early July, with a seminar scheduled for late July.

Table 1.1: MASc thesis requirements summary

Element	Requirement
Research Scope	Substantial problem extending beyond existing laboratory results
Literature Review	Comprehensive assessment of classical control and modern AI approaches
Validation	Both simulation (Gazebo) and hardware (Agilex) empirical evidence
Length/Format	Typically 80–120 pages; GSPA formatting and UWSpace submission
Committee	Supervisor plus two faculty readers; formal display period

1.5 Thesis Organisation

The remainder of this thesis is structured as follows. Chapter 2 reviews classical and learning-based control methods for articulated vehicles, alongside relevant perception and safety literature. Chapter 3 derives the state-space equations for the tractor-trailer system and analyses the jackknife instability. Chapter 4 presents the CNN and autoencoder-based perception stack. Chapter 5 defines the MDP formulation and details the DDPG and Decision Transformer implementations. Chapter 6 describes the Gazebo simulation environment and Agilex hardware integration. Chapter 7 presents and discusses experimental results. Chapter 8 summarises the findings and outlines future work.

Chapter 2

Literature Review

2.1 Articulated Vehicle Control

2.1.1 Classical Stabilisation Techniques

Classical articulated vehicle control methods address the hitch instability through structured model-based approaches:

- **PID control:** Simple to implement and computationally negligible, but incapable of anticipating the jackknife condition. Gains must be re-tuned for different payloads and path curvatures.
- **Model Predictive Control (MPC):** Optimises steering over a receding horizon subject to state and input constraints. Linear MPC can enforce hard bounds on the articulation angle γ .
- **Nonlinear MPC (NMPC):** Captures the full nonlinear kinematics including reverse instability. Projected to achieve cross-track errors near 0.35 m, but incurs 50–100 ms latency per control step.
- **Active hitch stabilisation:** Applies an additional actuated degree of freedom at the hitch to damp articulation oscillations. Research at Waterloo into trailer sway control and active hitches demonstrates that lateral articulation at the hitch can mitigate oscillations, but for a standard autonomous tractor-trailer this must be managed purely through the tractor’s steering angle δ .

2.1.2 Kinematic Models

The bicycle model provides a baseline for rigid-body longitudinal and lateral dynamics. For an articulated system, the state vector must be expanded to include the trailer heading and articulation angle, as detailed in Chapter 3.

2.2 Reinforcement Learning in Autonomous Driving

2.2.1 Policy Gradient Methods

Deep Deterministic Policy Gradient (DDPG) is an off-policy actor-critic algorithm designed for continuous action spaces. Prior work in ME 780 established that DDPG achieves a cross-track error of approximately 0.505 m for rigid-body path following, outperforming a PID baseline of 0.784 m on the same task.

Proximal Policy Optimisation (PPO) offers improved stability over DDPG through clipped probability ratios, but its on-policy sample efficiency is lower, which is a significant consideration when simulation rollouts are expensive.

2.2.2 Transformer-Based Methods

Decision Transformers (DT) reformulate RL as sequence modelling: an action is predicted from a context window of past states, actions, and returns-to-go using a causal self-attention mechanism. The ME 780 study found that DTs achieve a cross-track error of 1.717 m and are prone to oscillation if the training dataset is insufficiently diverse. To address this, the MASc research must utilise a dataset that includes recovery maneuvers, either by injecting noise into a DDPG expert’s actions or by explicitly including high-error-state trajectories.

2.3 Perception and Representation Learning

2.3.1 Convolutional Neural Networks for Driving

CNNs have become the standard front-end for autonomous driving perception, extracting spatial features from camera images and LIDAR-derived occupancy grids. The WATonoBus project at Waterloo utilises 3.2 MP Basler cameras and Robosense LIDARs to detect snow-covered curbs and traffic participants, providing an operational context directly relevant to this thesis.

2.3.2 Autoencoders for State Compression

Autoencoders compress high-dimensional observations into a compact latent representation, mitigating the curse of dimensionality for RL training. By pre-training an autoencoder on a diverse dataset of driving scenarios—such as the DriveSOTIF dataset—the encoder learns to capture essential geometric features of the road and obstacles. The RL agent then operates on the latent vector z , which reduces policy network parameters, accelerates convergence, and improves generalisation across environments.

2.4 Safety in Autonomous Systems

Safety of the Intended Functionality (SOTIF) defines the safety boundary for autonomous systems operating under nominal sensor and actuator conditions. For articulated vehicles, SOTIF analysis requires characterising the conditions under which the RL policy produces unsafe actions (e.g., allowing γ to approach the jackknife threshold). Control Barrier Functions (CBFs) offer a formal mechanism to overlay hard safety constraints on top of a learned policy, effectively bridging high-performance AI control with the deterministic safety requirements of mechatronics engineering.

2.5 Benchmarking Against Recent Waterloo Theses

To contextualise the expected level of rigor, Table 2.1 summarises recent MASc theses from the Mechatronics domain at the University of Waterloo.

Table 2.1: Selected recent MASc theses, University of Waterloo Mechatronics Engineering

Author (Year)	Topic	Methodology	Rigor Highlights
Joseph (2025)	Gaze-enabled grasping assistance	ROS2, Kinova Arm, Meta Quest Pro	User study with 30 participants; workload and performance metrics
Thibault (2022)	Humanoid robotic manipulation	EUROBENCH, REEM-C robot	Defined “loco-manipulation”; new manipulability–stability metrics
Hart (2023)	Gait detection via machine learning	Wearable sensors, ternary classification	6 classical + 1 Transformer model; grid-search hyperparameter optimisation
Wei (2021)	Path following for manipulators	Transpose Jacobian control	Free of inverse transformations; validated on 2-DoF and 4-DoF robots
Bickford (2019)	Autonomous bicycle stabilisation	Linear control, Whipple model	Multi-loop architecture; “virtual paths” for high-error recovery

Common themes include strong mathematical modelling foundations and comprehensive validation datasets. For this thesis, a simple “it works” result is insufficient; the

candidate must quantify stability limits, evaluate performance across varied paths, and compare against classical benchmarks.

Chapter 3

Mathematical Modeling and Problem Formulation

3.1 Tractor Dynamics

The tractor longitudinal and lateral dynamics are:

$$\dot{x} = v_x \cos \psi - v_y \sin \psi \quad (3.1)$$

$$\dot{y} = v_x \sin \psi + v_y \cos \psi \quad (3.2)$$

where v_x is the longitudinal velocity, v_y is the lateral velocity, and ψ is the vehicle heading.

3.2 Trailer Articulation and Reverse Instability

The articulated system introduces hitch geometry defined by:

- L_h : distance from tractor rear axle to hitch point
- L_t : distance from hitch point to trailer axle
- γ : articulation angle between tractor and trailer

The hitch angle kinematics during reverse motion are:

$$\dot{\gamma} = \frac{v}{L_t} \sin \gamma - \frac{v}{L_h} \tan \delta \cos \gamma \quad (3.3)$$

Reverse motion introduces instability as γ approaches the jackknife threshold.

3.3 Challenges of Reverse Maneuvering

Reverse motion is characterised by a sign reversal in the kinematic equations, which transforms the stabilising effect of forward velocity into a destabilising force. Numerical experiments using NMPC have shown that successful reverse maneuvering requires optimisation of the steering input over a prediction horizon to prevent γ from reaching its mechanical limits. The RL agent developed in this thesis must learn analogous foresight: rewarding actions that keep the trailer aligned with the target path while maintaining γ within a safe operational envelope.

3.4 MDP Formulation

The control problem is cast as a Markov Decision Process (MDP) $(\mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \gamma_d)$ where:

- $\mathcal{S}$: observation space consisting of a vector of state variables (steering angle, articulation angle, cross-track error, heading error) and an 84×84 grayscale BEV image;
- $\mathcal{A}$: continuous action space $[\dot{\delta}, v] \in [-15^\circ/\text{s}, 15^\circ/\text{s}] \times [0, 2 \text{ m/s}]$;
- $\mathcal{T}$: transition dynamics simulated in Gazebo;
- $\mathcal{R}$: reward function defined in Chapter 5;
- γ_d : discount factor.

Episodes terminate on collision, jackknife ($|\gamma| > 90^\circ$), or successful path completion.

Chapter 4

Perception and Observation Space Design

4.1 CNN Encoder

A Convolutional Neural Network serves as the front-end of the RL agent, processing raw visual inputs or LIDAR-derived occupancy grids. This allows the agent to perceive the environment directly, facilitating lane detection and obstacle avoidance without requiring a pre-defined symbolic representation of every object. The WATonoBus project utilises a similar multi-modal sensor fusion approach, combining 3.2 MP Basler cameras and Robosense LIDARs to detect snow-covered curbs and traffic participants.

The CNN architecture extracts spatial features from:

- occupancy grids derived from LIDAR point clouds
- camera images (lane markings, obstacles)
- lane boundary segmentation masks
- BEV projections of the surrounding environment

These features are passed to the RL policy network. The “curse of dimensionality” is managed by focusing the CNN’s receptive field on the area immediately surrounding the vehicle and its projected path. The CNN is trained in tandem with the DDPG actor, allowing the system to learn which visual features are most relevant for path maintenance and collision avoidance.

4.2 Autoencoder Latent Space

To improve RL training efficiency, an autoencoder compresses high-dimensional sensor observations into a compact latent representation. The encoder $E(\cdot)$ maps the observation

s to a latent vector z , and the decoder $D(\cdot)$ reconstructs the original observation:

$$z = E(s) \quad (4.1)$$

$$\hat{s} = D(z) \quad (4.2)$$

By pre-training the autoencoder on a diverse dataset of driving scenarios—such as the DriveSOTIF dataset—the encoder learns to capture essential geometric features of the road and obstacles in a low-dimensional format. The RL agent then operates on z rather than the raw observation, which reduces the number of parameters the policy network must learn, leading to faster convergence and better generalisation across environments.

4.3 Sensor Fusion

LIDAR point clouds are processed via ROS2 point cloud libraries and converted to 2D occupancy grids before being passed to the CNN. Camera data provides complementary lane boundary information. A collision module based on Time-to-Collision (TTC) provides a predictive safety signal that penalises imminent impacts.

Table 4.1: Perception stack components and their roles

Component	Role	Implementation
CNN Encoder	Feature extraction from occupancy grids and camera feeds	Integrated as initial layers of the DDPG actor-critic network
Autoencoder	Dimensionality reduction of complex environmental observations	Pre-trained on Gazebo-generated datasets to establish latent state z
LIDAR Fusion	Obstacle and lane boundary detection in 3D space	Processed via ROS2 point cloud libraries; converted to 2D grids
Collision Module	Predictive safety layer to penalise imminent impacts	Based on Time-to-Collision (TTC) and safety stop requirements

Chapter 5

Reinforcement Learning Framework

5.1 Deep Deterministic Policy Gradient (DDPG)

DDPG is an off-policy actor-critic algorithm specifically designed for continuous action spaces, making it well suited to steering and acceleration control. The actor network $\mu(s \mid \theta^\mu)$ maps states to deterministic actions, while the critic network $Q(s, a \mid \theta^Q)$ evaluates action quality via the Bellman equation:

$$Q(s_t, a_t) = r_t + \gamma_d Q(s_{t+1}, \mu(s_{t+1})) \quad (5.1)$$

Parameters are updated by minimising the temporal difference error in the critic and maximising the expected return through the actor using the deterministic policy gradient theorem.

5.1.1 Reward Function

The proposed reward function is:

$$R = w_1 e^{-|e_y|} + w_2 e^{-|e_\psi|} - w_3 |\gamma| - w_4 C \quad (5.2)$$

where:

- e_y = trailer cross-track error
- e_ψ = heading error
- γ = articulation angle
- C = collision penalty

A multiplicative alternative is also considered:

$$R = e^{-|e_y|} \cdot e^{-|e_\psi|} \quad (5.3)$$

5.1.2 Multiplicative vs. Additive Reward Structures

Prior results indicated that a multiplicative reward structure forces the agent to minimise both cross-track and heading errors simultaneously, preventing the policy from exploiting partial credit in the additive form. In this thesis, this logic is extended to include hitch articulation stability: the multiplicative penalty on $|\gamma|$ discourages the agent from trading path accuracy for a large articulation angle.

5.2 Decision Transformer

The Decision Transformer (DT) reformulates RL as a sequence-to-sequence problem. Given a context window of past states $s_{t-k:t}$, actions $a_{t-k:t-1}$, and returns-to-go $\hat{R}_{t-k:t}$, the model predicts the next action using the causal self-attention mechanism of the Transformer architecture:

$$a_t = \pi_\theta(s_{t-k:t}, a_{t-k:t-1}, \hat{R}_{t-k:t}) \quad (5.4)$$

This architecture is particularly suited to reverse maneuvering, where an early steering error may only result in a jackknife several seconds later, requiring the model to reason over long-term dependencies.

5.2.1 Dataset Requirements

The ME 780 study revealed that DTs are prone to oscillation if the training dataset is too narrow. To address this, training data must include recovery manoeuvres obtained by:

- injecting noise into a DDPG expert’s actions during data collection;
- explicitly including trajectories initialised from high-error states (e.g., $|e_y| > 1$ m or $|\gamma| > 20^\circ$);
- including step discontinuities in the reference path to force the model to learn re-acquisition behaviour.

5.3 Hyperparameter Optimisation

Both the DDPG and DT agents require careful hyperparameter tuning. The actor-critic learning rates, replay buffer size, target network update rate, and noise scale are selected via grid search over simulation rollouts. Full hyperparameter tables are provided in Appendix B.

Chapter 6

Simulation and Experimental Implementation

6.1 Gazebo Simulation Environment

Gazebo serves as the primary training environment. Unlike simple 2D simulators, Gazebo uses a physics engine to model the inertial properties, tire friction, and mechanical constraints of both the tractor and trailer. This is essential for simulating the jackknife condition, allowing the agent to experience the physical consequences of unstable reversing manoeuvres without risking damage to the Agilex hardware.

The simulation environment includes:

1. **URDF Modelling:** A detailed Unified Robot Description Format file for the Agilex tractor and its custom-built trailer, including the revolute joint at the hitch point.
2. **Obstacle Generation:** Randomly placed static and dynamic obstacles to train the obstacle avoidance module.
3. **Lane Markings:** Visual representations of lane boundaries to facilitate training of the CNN-based lane detection module.
4. **Collision Detection:** Contact sensors triggering episode termination and penalty on collision.
5. **Noisy Sensor Simulation:** Gaussian noise on LIDAR returns and camera images to model adverse weather conditions relevant to SOTIF analysis.

6.2 Autoware and ROS2 Integration

Autoware is an open-source software stack for self-driving vehicles providing modules for localisation, sensing, and planning. The RL agent interfaces with ROS2 nodes, subscribing to sensor data (LIDAR, cameras) and publishing steering and velocity commands to the Agilex motor controllers.

The Waterloo MVS lab emphasises high-definition (HD) maps and robust localisation to provide a consistent reference frame. The HD map, generated from high-precision LIDAR data, allows the RL agent to determine its position relative to lane boundaries with sub-centimetre accuracy, serving as the global path reference.

6.3 Hardware Platform

Experimental validation is performed on:

- **Agilex robotic platform:** The physical tractor unit, equipped with Robosense LIDAR and Basler cameras consistent with the WATonoBus sensor suite.
- **Custom trailer:** A purpose-built trailer with a passive hitch joint, replicating the kinematic constraints of the simulated system.
- **Sim-to-Real transfer:** Domain randomisation in Gazebo (varying friction coefficients, obstacle configurations, lighting) is used to reduce the reality gap.

6.4 Testing Protocols

6.4.1 Forward Motion and Obstacle Avoidance

Table 6.1: Forward motion experiment definitions

Experiment		Objective	Success Metric
Path Fidelity		Maintain low CTE on varying curvature paths	Mean Absolute CTE < 0.2 m
Dynamic Avoidance	Obstacle	Divert from path to avoid moving actors and return safely	Zero collisions; minimum separation distance maintained
Lane Detection Accuracy		Identify lane boundaries via CNN under varied lighting	F1-score of lane segmentation mask

6.4.2 Reverse Maneuvering and Jackknife Prevention

This is the most technically challenging test series. The research quantifies the “Safe Articulation Zone” and tests the agent’s ability to remain within it during complex backing tasks:

1. **Straight-line backing:** Reverse 20 m without the articulation angle exceeding 5° .
2. **90-degree docking:** Reverse the trailer into a bay with width only 20% greater than the trailer width.
3. **Jackknife recovery:** Initialise with $|\gamma| = 30^\circ$ (near-critical); recover to $|\gamma| < 5^\circ$ within 5 m of travel.

6.4.3 Perception and Sensor Degradation

The CNN/autoencoder observation space enables robustness testing under simulated sensor degradation. Gazebo is configured to reduce image contrast and add LIDAR noise (simulating adverse weather), and the resulting increase in cross-track error is measured. This provides SOTIF-relevant insight into the operational design domain boundaries of the system.

Chapter 7

Results and Discussion

7.1 Path Following Performance

7.2 Reverse Maneuvering Success

7.3 Perception Robustness

7.4 Benchmark Comparison

Table 7.1: Controller performance comparison

Model	Avg. CTE (m)	Jackknife Events/100	Latency (ms)
PID	0.784	High	< 5
NMPC	0.350	Low	50–100
DDPG	0.505	Medium	10–20
Decision Transformer	1.717	High	30–50

The results must be understood in terms of multi-objective performance: in rigid-body systems, success is synonymous with minimising cross-track error. In articulated systems, hitch stability takes precedence over exact path following. An agent that maintains $|\gamma| < 5^\circ$ throughout a reverse task is more successful than one that achieves lower CTE but triggers jackknife events.

Table 7.2: Experiment matrix: tasks, observations, and success criteria

Experiment	Task	Observation	Success Metric
Path fidelity (forward)	Path following	BEV image + vector state	Mean Abs. CTE < 0.2 m
Dynamic obstacle avoidance	Avoidance	BEV image + vector state	Zero collisions
Lane detection	Perception	Camera image	F1-score of lane mask
Straight-line backing	Reverse	BEV image + vector state	$ \gamma < 5^\circ$ over 20 m
90-degree docking	Reverse	BEV image + vector state	Trailer in bay; no collision
Jackknife recovery	Stability	Vector state	Recover to $ \gamma < 5^\circ$ in 5 m
Sensor degradation	Robustness	Noisy BEV image	CTE increase < 50% vs. clean

Chapter 8

Conclusion

8.1 Summary of Findings

This thesis presented a reinforcement learning framework for articulated autonomous vehicle control. The key contributions demonstrated are:

- forward and reverse maneuvering with active hitch stability control;
- perception-aware obstacle avoidance via CNN-based occupancy grid processing;
- jackknife prevention through reward engineering and articulation angle penalties;
- sim-to-real deployment on the Agilex platform via ROS2 and Autoware.

The integration of vision-based perception addresses the “black box” nature of earlier RL systems. By visualising the attention masks of the CNN encoder or the reconstruction accuracy of the autoencoder, it is possible to gain interpretable insight into the steering decisions of the policy. This interpretability is essential for eventual deployment in public environments such as the University of Waterloo campus, where the WATonoBus operates.

8.2 Limitations

The study is subject to the following constraints:

- Experiments are conducted within a specific Operational Design Domain (ODD); performance outside these conditions is not characterised.
- Sensor noise models in Gazebo approximate but do not fully replicate all real-world degradation modes (e.g., lens icing, LIDAR blooming).
- The current framework assumes a fixed payload; hitch dynamics change significantly with varying trailer mass.

8.3 Future Work

Several directions remain open for future investigation:

- **Multi-trailer systems:** Extending the kinematic model and RL policy to double or B-train configurations.
- **Online adaptation:** Updating the policy in real time to compensate for varying payloads and road conditions.
- **Control Barrier Function overlays:** Implementing CBFs as a hard safety layer that overrides the RL agent when a collision or jackknife is imminent, bridging high-performance AI control with the deterministic safety requirements of mechatronics engineering.
- **Formal verification:** Applying reachability analysis to provide provable bounds on the articulation angle under the learned policy.
- **Multi-modal perception:** Incorporating radar and thermal imaging to improve perception robustness in adverse weather.

Appendix A

ROS2 Software Stack

Appendix B

Full Hyperparameter Tables

Table B.1: TD3 Hyperparameters

Hyperparameter	Value
Learning rate	
Batch size	
Replay buffer size	
Policy noise	
Target update rate	
Timesteps	200,000

Appendix C

Additional Plots

Appendix D

Reward Function Derivation

Bibliography